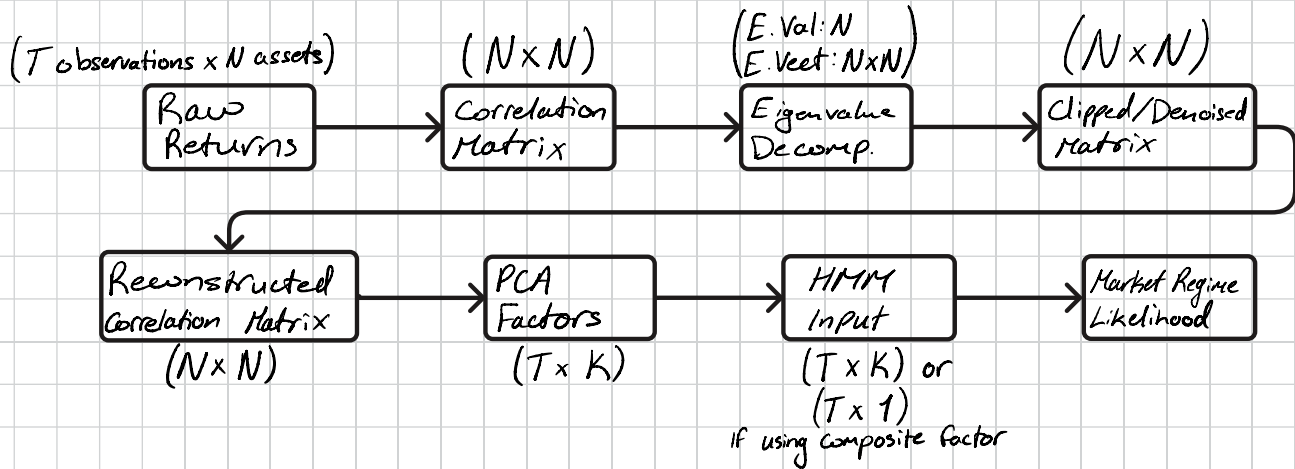




Data Flow Diagram



Variable Decisions

PCA:

Will use 3 PCs

→ Need PC2 & PC3 $\perp$ PC1 & PC2 $\perp$ PC3 $\Rightarrow$ uncorrelated patterns

→ will be checked by examining corr. matrix

HMM:

Will fit 3 states, Bull/Low-vol Regimes, Bearish/high-vol Regimes, Neutral/transitional regimes.

RMT:

σ^2 - fitted to optimized val by least square method

$q = \frac{m}{n}$, where m is # of variables/asset & n is # of observations/time step

Grid log-likelihood:

HMM:

- Fit separate HMMs w/ 2, 3, 4 states, then compute each models log-likelihood on the same data
- Use BIC instead of raw log-likelihood. $BIC = -2 \ln(\hat{L}) + k \ln(n)$
→ $\hat{L}$ = Maximized likelihood, k = # of free parameters, n = # of observations
Lower BIC = better

for n -states in $[2, 3, 4]$:

model = fit_hmm(factors, n -states)

ll = forward_algorithm(model, factors)

bic = $-2 * ll + n_params(n_states) * \log(T)$

pick n -states w/ lowest BIC

Marchenko-Pastur q & σ^2 :

- Search separate q & σ^2 , fit theoretical MP density each time
- Pick pairs that minimize distance btwn theoretical MP density & empirical e.val histogram

Validation Checkpoints

- Code as automated checks (unit tests/assertions) when:
 - Rerun everytime data & parameters are changed, or pipeline re-runs

Ex:

- Is the denoised corr. matrix +ve semi-def.
 - Do e.vals sum to the trace
 - Are all HMM transition matrix rows summing to 1
 - Did log-likelihood inc. or stay the same at each Baum-Welch iteration
- (Should be assert statements or small test functions)

- Manual/Visual inspection when:

Ex:

- Do the e.val spectrum plots look reasonable compared to Marchenko-Pastur density (MP)
 - Do the decoded Bull/Bear/Neutral regimes line up w/ known market history (2008, 2020, 2022)
 - Does the HMM regime timeline look economically sensible
- (Best done by plotting/reviewing)

Pseudocode

```
build_correlation_matrix(returns) → corr_matrix  
fit_marchenko_pastur(corr_matrix) → (q, sigma2, lambda_plus)  
denoise_correlation(corr_matrix, lambda_plus) → denoised_corr  
run_pca(denoised_corr, k) → factors, loadings  
fit_hmm(factors, n_states) → model  
decode_regimes(model, factors) → regime_labels
```


4. Reconstruction step after clipping

You have the clipping procedure from earlier notes, but this sketch doesn't explicitly restate how you'll rebuild a valid correlation matrix after replacing the noisy eigenvalues (i.e., $\hat{C} = V \Lambda_{\text{clipped}} V^T$, then rescaling the diagonal back to 1s so it's a proper correlation matrix, not just a PSD matrix). This diagonal-rescaling step is often skipped and causes downstream bugs.

5. Baseline comparison structure

Since your paper's core contribution is "denoised vs. raw," the sketch should explicitly show two parallel branches: one pipeline using raw correlation matrices straight into PCA/HMM, and one using the denoised version — with the same validation and grid search logic applied to both, so your comparison is apples-to-apples.

6. Output/results specification

Not yet stated: what exactly gets saved/plotted as your final artifacts — e.g., regime timeline plots, transition matrix heatmaps, BIC comparison table, eigenvalue spectrum overlay plots. Deciding this now avoids scrambling for figures later.

Suggested next addition

Add one more page/section to your sketch: an end-to-end pipeline diagram showing the raw and denoised branches side by side, each annotated with matrix shapes, followed by a short list of function signatures. That would make this sketch fully complete and ready to translate directly into Python modules.

Would you like me to draft that missing end-to-end diagram and function-signature list now, using your existing notation and decisions?